

Passion or Public Disorder:

*Exploring Philadelphia Police Stop Data in Relation to Major Sporting
Events in the City*

**Medhavi Jambhekar, Vijetha Ramdas, Charlotte Suarez,
Jasmine Xu, Alejandro Zamudio**

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Introduction

Background

The city of Philadelphia has earned a notorious reputation in the sports world due to the actions of its fans. Back in 1989, Philadelphia Eagles fans infamously booed Santa Claus and “pelted players, refs, and broadcasters with snowballs” (Gregory n.p.). At the Eagle’s previous stadium, unruly fan behavior was so out of hand that a working “jail and courtroom were installed,” under the stadium (Gregory n.p.). In 2011, GQ magazine ranked the Philadelphia Phillies and Eagles as having the first and second in a list of “The Worst Sports Fans in America” (Winer n.p.). Philadelphia fans are also no strangers to true violence, either. In 2012, surveillance cameras caught three people wearing Flyers jerseys attacking fans wearing opposing teams' jerseys in a restaurant in South Philadelphia (Stamm n.p.). Following the Eagles' win in the 2025 Super Bowl, fans took to the streets of the city, tearing down light poles and setting off fireworks in the middle of the street (Simon and Sanders n.p.). As a result, the media often oscillates between descriptive words like “passionate” and “destructive” when referring to the sports culture that has arisen in Philadelphia. An expectation has been set, and reinforced time and time again, that “Win or lose, Philly burns.”

Despite the prominence of this stereotype, it remains unclear whether such portrayals reflect measurable changes in real-world behavior. The extent to which major sporting events influence Philadelphia’s law enforcement activity (such as traffic stops or pedestrian stops) is an area of interest. Given the intensity with which sporting events are portrayed, understanding whether major sporting events shape policing patterns and the frequency of vehicle or pedestrian stops made on these particular days is both socially and analytically significant.

Objectives

This project aims to investigate whether the “Philly will burn” stereotype is supported by empirical evidence, specifically found within police stop data. This study is guided by two primary objectives:

1. Comparison of Stop Patterns

Examine how traffic and pedestrian stop patterns differ between game days and non-game days across major league sports in Philadelphia. This will include analyzing stop frequency, arrest rates, and other stop outcomes, as well as observing variation across sports and proximity to stadium locations.

2. Predictive Modeling of Game Day Outcomes

Determine whether game-related characteristics can predict traffic stops or arrest rates by analyzing the extent to which score differential, the opponent played, or game significance affect the nature of stops performed on a game day. For example, is it reasonable to predict that sports fans will be more animated when their team is playing their biggest rival, such as when the Eagles play the Dallas Cowboys? This study aims to identify how these characteristics influence predictions of traffic stop outcomes.

Ultimately, this project seeks to assess whether widely held perceptions of Philadelphia sports culture are grounded in real data or driven primarily by anecdotal evidence.

Literature Review

This project builds upon an established body of research examining the relationship between spectator sports and crime. Carter, Moule, and Knode analyze crime patterns in 21 U.S. cities, comparing incidents during home and away games for baseball and football teams. With a

focus on how far from the stadium criminal activities are concentrated, the research uses a Quasi-Poisson regression and aggregates crime data in three distance ranges, ultimately finding that certain types of crimes occur more frequently in areas closer to stadiums on game days (Carter et al. n.p.).

Additionally, Bowman investigates arrests within football stadiums, focusing on how game-related characteristics influence arrest rates. Using arrest data acquired from requests to various police departments, Bowman uses various regression models to analyze differences in arrests and tie characteristics – such as whether the team won, betting odds, and importance of the game – to note slightly higher arrest outcomes (Bowman n.p.). While Bowman’s study focuses primarily on “in-stadium arrests,” his process of evaluating occurrences of crime in those stadiums is helpful in forming the methods in this study.

Furthermore, both studies cover how they addressed key confounders such as the time of day, the general location of the event, and the prevalence of law enforcement within areas of the stadium. While Bowman’s emphasis on study is centered on “in-stadium arrests”, both reference Routine Activities Theory as an explanation for crime occurring near or within event venues (Bowman; Stamm). Ultimately, their insight on these issues and confounders can inform the analysis steps in this study and how the results will be interpreted in context.

Building on prior research, this study narrows its focus to the city of Philadelphia. While previous work, such as Carter’s study, examines similarities and differences among multiple cities in the United States, this study provides a detailed analysis of the Philadelphia area, infamous for its passionate sports fanbase. Furthermore, this study includes all four major professional sports teams in the city (football, basketball, baseball, and hockey), extending beyond prior literature that typically focused on one or two sports. All venues are within the

same area of the city, enabling the exploration of the individual impact each sport has along with the combined effects of multiple sporting events in one day.

Data Analysis Pipeline

This study will begin with the collection and cleaning of the data. While the main traffic stop dataset will come from the Stanford Open Policing Project, all sports datasets will be collected from separate sources. The goal of the collection and cleaning process is to create standardized datasets for the traffic stops and all four sports schedules that can be readily used for initial analysis and data exploration.

Next, Data Wrangling will be performed to aggregate stop data and join it with key information from the game day datasets, including which sports were played on those days, whether the observation took place before or after the game, and the game results. The study will then move into the Data Exploration phase, where important characteristics of the sporting and policing data will be visualized, and a spatial heatmap will be created to understand the frequency of stops around Philadelphia.

Non-numeric variables will be encoded as needed, and data will be split into training and testing sets to prepare for modeling. This study will create a separate machine learning model for each sport to predict the number of police stops on game days using the game characteristics as features. Various candidate models will be fit on the training data and selected based on performance on the validation set. Final model performance metrics will be calculated using an unseen test dataset. Finally, model interpretability techniques will be utilized to understand the contribution of each feature.

All findings will be communicated by presenting the results of the analysis and model, and tying them back to the set project objectives for this study.

Data Description

The main dataset for this study is the Philadelphia Traffic and Pedestrian Stop data from the Stanford Open Policing Project. Containing data from 2014 to 2018, each observation represents a traffic or pedestrian stop performed by an officer in Philadelphia. The attributes associated with each stop include date, time, and location, driver characteristics such as race, sex, and age, and details about the stop, including whether contraband was found or an arrest was made. Containing over a million observations, the dataset will serve to highlight where stops are being made, how often they are occurring, and what characteristics are common among these stops. Furthermore, the location and time information can be used to merge with the sports related data and identify prevalence near venues.

For the supplemental data, this study will utilize schedule and score data from all four major sports teams in Philadelphia. This includes the Eagles (football), the 76ers (basketball), the Phillies (baseball), and the Flyers (hockey). Each team's dataset includes details such as the date the game was played, where the game was played (home/away), the score of the game, and the opponent. The supplemental data will be used to filter or stratify the police stop data based on when major sports games occurred in Philadelphia and to explore patterns between game outcome and police stop patterns. All sports data will be downloaded from the *Sports Reference* site associated with each sport. *Sports Reference* is a set of websites that collect schedule, game, player, and historical data for major league sports. Each sports-specific site contains schedule and result data by season, such as the 2014 National Football League season. For each sport,

individual season schedules were exported and merged to form singular datasets associated with the sport. This resulted in four datasets, one for each sport, including game result data for the five-year span in which the traffic data occurred.

Data Cleaning and Wrangling

Observations in the policing dataset with missing date, time, or location information were removed, as these attributes are essential to this study. For the sports datasets, missing values in variables such as game start time or attendance were researched in online sources and manually recorded. Variables irrelevant to the research objectives, such as certain game or player-specific statistics unlikely to meaningfully impact fan behavior or policing outcomes, were removed from the data. Date columns across all datasets were standardized to a consistent format to prevent errors during analysis. Additionally, all Boolean variables were encoded as 0 and 1 to ensure compatibility with machine learning models. The result of this stage was a set of cleaned datasets containing only relevant variables and no missing values.

The first step in data wrangling was grouping the stops into 1-kilometer square regions. This size was selected because it is easily interpretable and reasonably small to capture the outline of Philadelphia. To create the regions, the latitude and longitude coordinates of each stop in the police data were converted to kilometer-based coordinates and rounded down to the nearest kilometer. These coordinates were combined to create a region identifier for each stop. Figure 1 shows the defined regions overlaid on a map of Philadelphia.

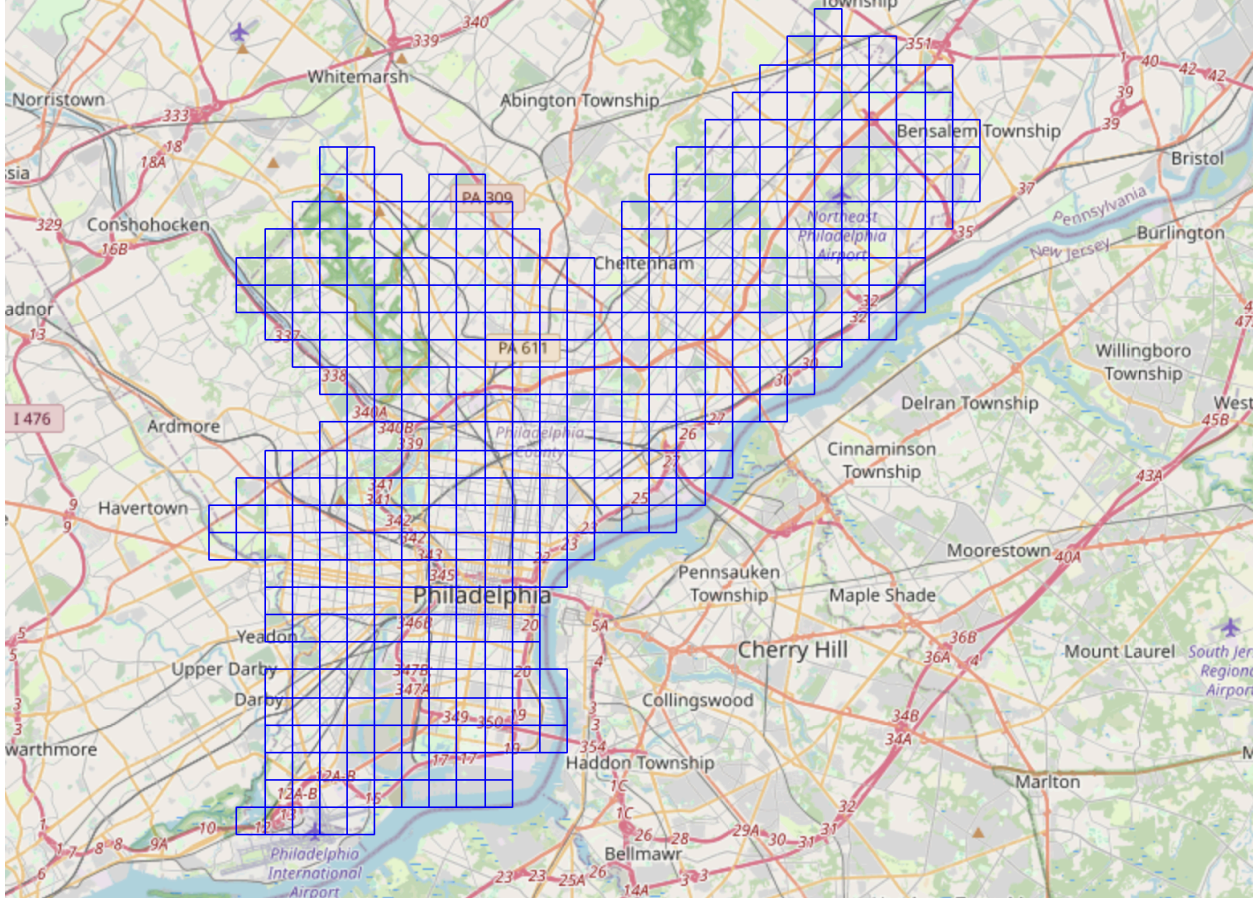


Figure 1. Map of Philadelphia with defined 1-kilometer square regions.

Next, the police stop data were aggregated by region identifier, date, and hour, and the total number of police stops was computed for each combination. Because each row in the original data represents a single stop, the resulting count variable represents the number of stops that occurred within a given region, day, and hour. Missing combinations correspond to periods with no stops and were therefore imputed with a stop count of zero.

For each of the four sports datasets, additional features were engineered to capture critical sport-related context that may influence fan behavior. These variables included indicators of whether the opponent was a notable rival, whether the game extended beyond regulation time (e.g., overtime or extra innings), and whether the game was a regular season or a playoff game

(e.g., quarterfinal, semifinal, or finals). Lastly, game dates were transformed into a day-of-the-week variable.

The aggregated police stop data was left-joined with each of the sport datasets separately to create four merged datasets. All subsequent data wrangling steps were performed independently for each dataset. A game-day indicator variable was constructed to distinguish between game days and non-game days. To create a dataset for modeling, all non-game day observations were removed to ensure a consistent set of predictors across all variables. Game start and end times were rounded to the nearest hour. Using these variables, a categorical game period variable was created to classify each observation as occurring before, during, or after the game.

Data Exploration

After the data wrangling, visualizations were created to understand patterns in both the game day and police stop data, which will assist in answering the first project objective.

Figure 2 compares the number of stops on non-game days against that of each of the four sports' game days. The dataset was aggregated on the date and hour, and a stop count was computed for each (date, hour) group. Each sports' boxplot is based solely on days where the only game in Philadelphia on that date was for that sport, so it does not take into account days where games were played across multiple sports. From the plot, we can see that there is no significant difference in police stop frequency across sports, as well as between game days and non-game days.

Table 1 shows the number of observations, average, and median stops per (Date, Hour) group, the number of unique days represented in each category, and the total number of stops. It

is worth noting that across the six categories, the No Game category has the largest number of observations, while the Phillies and the Eagles have the most and least game days out of the four sports categories, respectively. This should be taken into account when observing the error bars present in Figures 3 and 4.

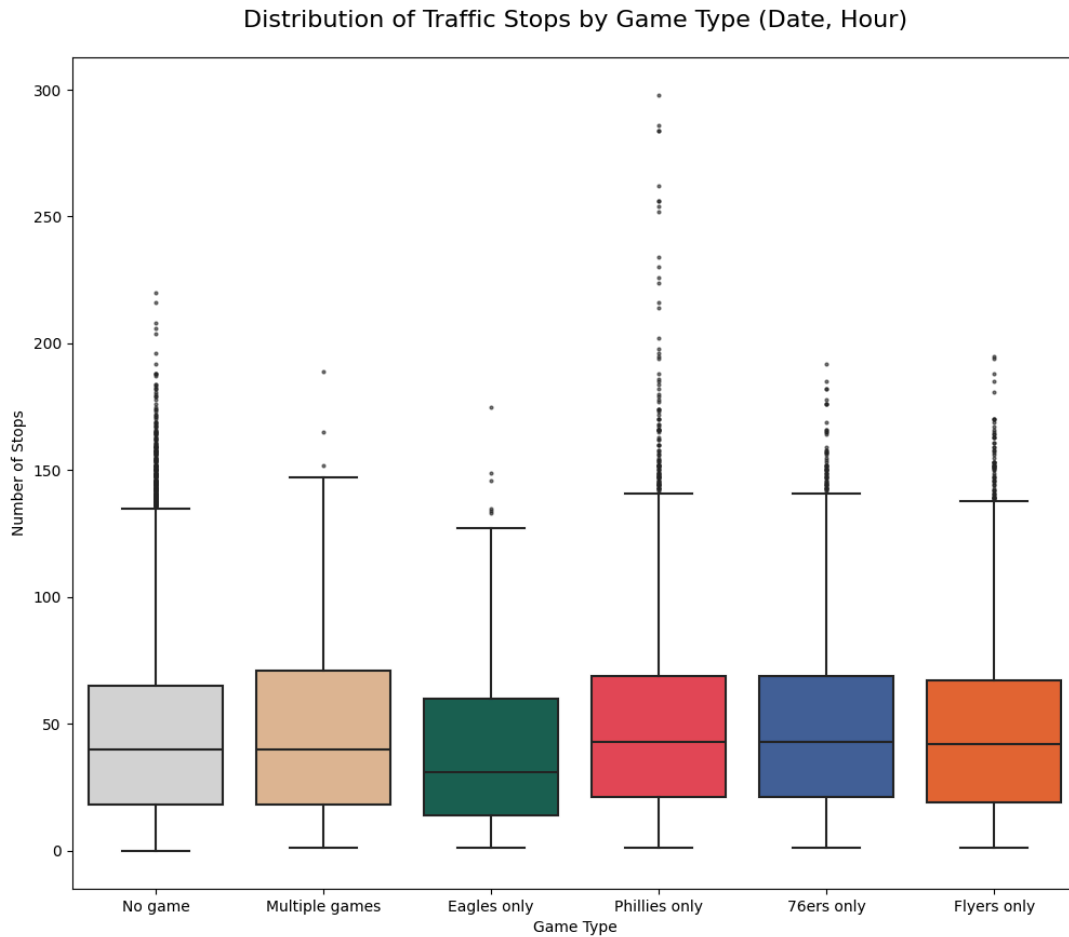


Figure 2. Boxplots comparing the number of stops on game days and non-game days.

Sport	Average Stops per (Date, Hour)	Median Stops per (Date, Hour)	Number of Unique Days	Number of Observations	Total Stops
No Game	45.897	40	927	21,026	965,039
Multiple Games	46.987	40	43	1,014	47,645
Eagles Only	41.206	31	26	622	25,630

Phillies Only	49.561	43	303	7,244	359,022
76ers Only	48.831	43	161	3,820	186,536
Flyers Only	48.015	42	161	3,834	184,088

Table 1. Table showing the number of observations in each sport grouping.

Figure 3 shows barplots that display the percentage of stops that result in an arrest or contraband found across all sports' game days. It appears that the arrest rates on Eagles and Phillies game days is higher than that of other game day types, though these differences are small, and do not hold across both barplots. The plots generally suggest that the rates of arrests and contraband found are not significantly impacted by the occurrence of a major sporting event. From Table 1, it is worth noting that the variance across the 6 categories of gamedays is large - there are 21,026 observations on non-game days, whereas within the game days, the Eagles have the least number of observations and the Phillies have the most. This is due to the fact that the Eagles only play 8 home games per regular season, while the Phillies play 81.

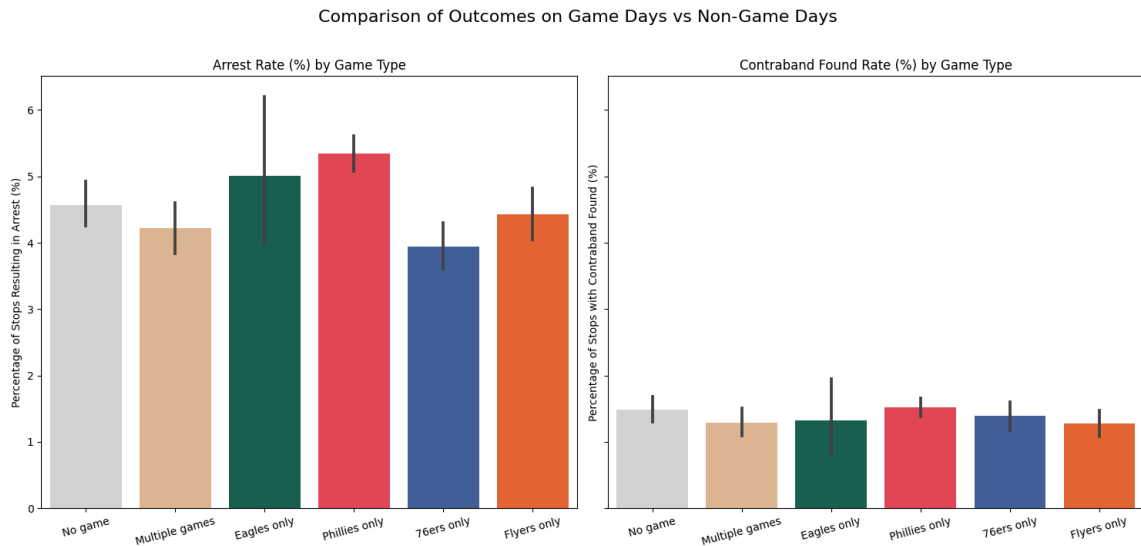


Figure 3. Barplots comparing arrest and contraband rates on game days and non-game days.

Figure 4 is a line graph that analyzes the average number of stops at each hour of the day across each game day type. We can observe that there tends to be a jump in arrests in the evening hours, especially around 6 PM (18:00). This is consistent with the most common game times for all major sporting events being between 6 and 8 PM. However, this pattern may also reflect unrelated temporal trends in community or police activity.

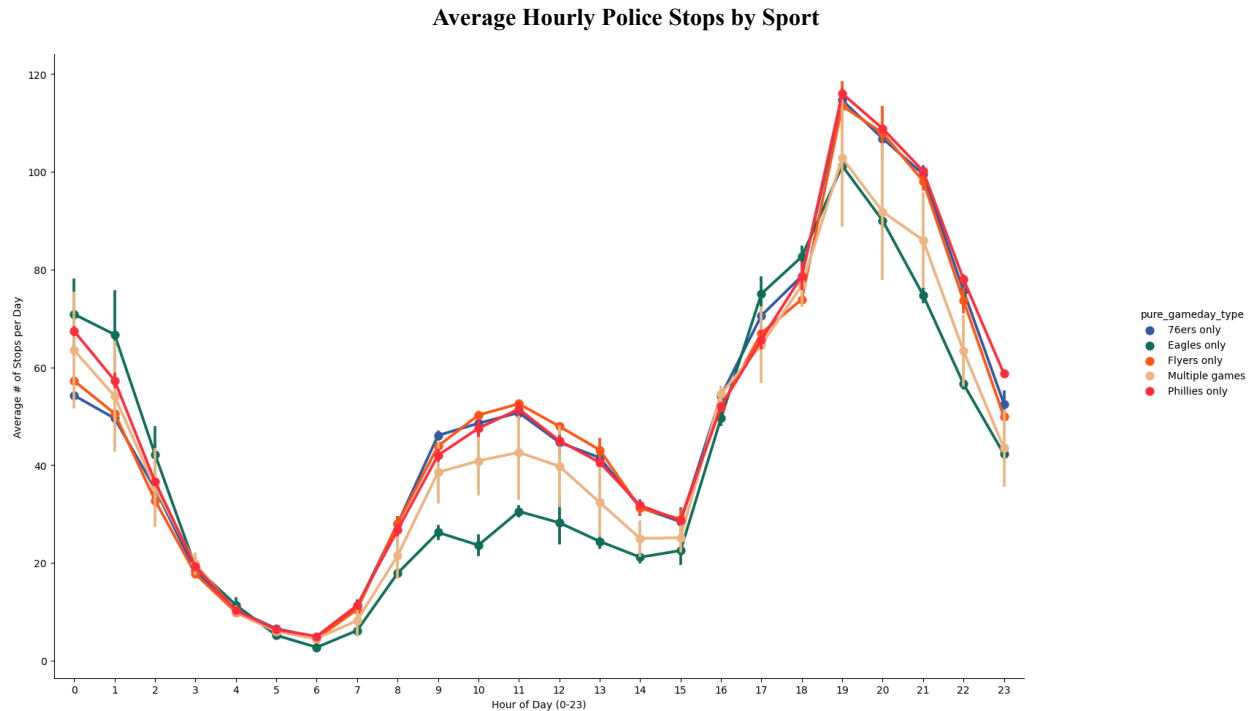


Figure 4. Line graph showing hourly distribution of stops across game days and non-game days.

Figure 5 is a spatial heatmap of the average hourly police stop frequency by region on Eagles' game days. As shown in the legends, darker squares indicate a higher average number of stops per hour than lighter squares. The green circle marks the location of the Eagles stadium. There are clear areas in the city that have higher police stop frequency. Notably, none of these areas are within close proximity to the Eagles stadium. This highlights a potential limitation in the data, as stadiums might hire private security for which we do not have data.

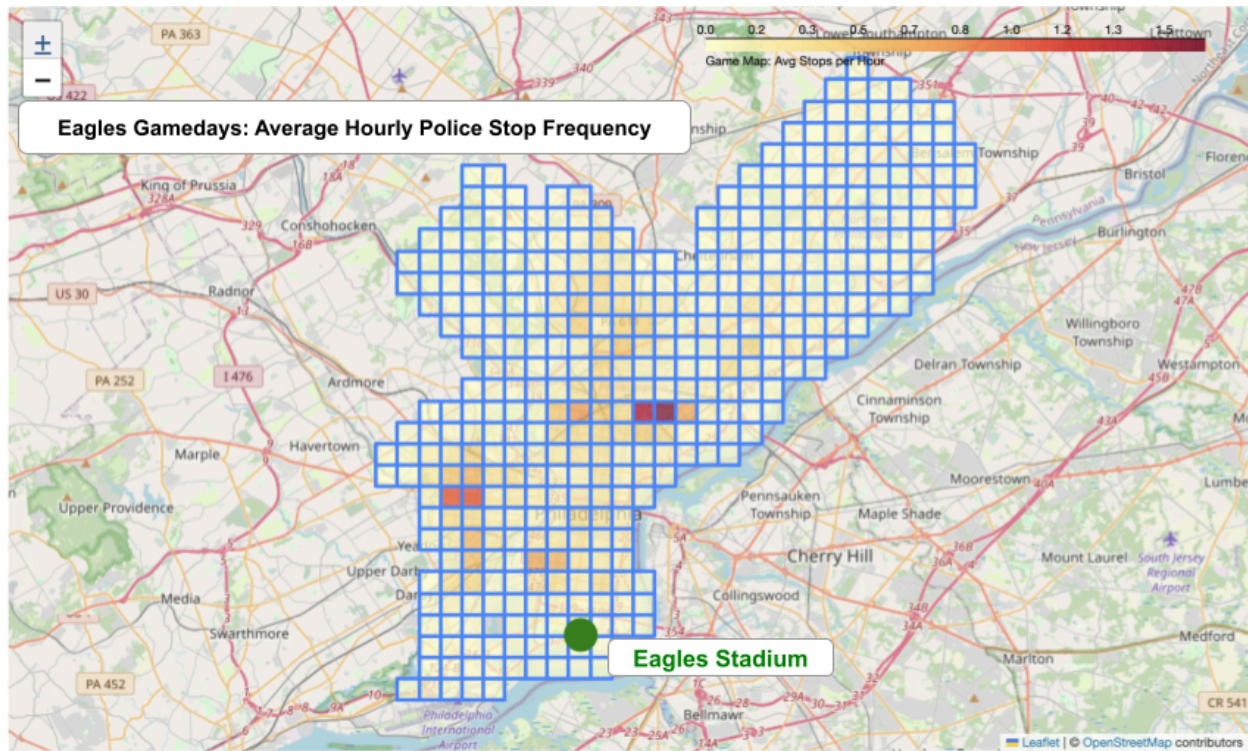


Figure 5. Spatial heatmap with average hourly stop rate per region on Eagles game days.

Modeling

Modeling Data Preparation

To address the second project objective, machine learning models were developed to predict the number of police stops per region and hour on game days. Separate models were constructed for each Philadelphia team because recorded game statistics differ across sports. The target variable for all models is defined as the number of police stops occurring within a given day, hour, and region. All predictor variables are described in Table 2, along with their corresponding data types. The “Sport” column indicates whether a variable is available across all four sports or is specific to a particular sport.

Variable	Description	Type	Sport
hour	Hour of observation	Numeric	All
Day	Day of week	Categorical	All
Home	Home game indicator	Boolean	All
PointDiff	Team score minus opponent score	Numeric	All
Win	Philadelphia win indicator	Boolean	All
Game	Game occurrence indicator	Boolean	All
Attendance	Game attendance	Numeric	All
Opp	Opponent team	Categorical	All
TeamScore	Philadelphia team score	Numeric	All
OT	Overtime indicator	Boolean	Flyers, 76ers, Eagles
SO	Shootout indicator	Boolean	Flyers
Season	Season identifier	Categorical	All
Wins	Season wins	Numeric	Flyers, 76ers, Eagles
Losses	Season losses	Numeric	Flyers, 76ers, Eagles
Streak	Current win/loss streak	Numeric	All
Rivalry	Rivalry game indicator	Boolean	All
Regular Season	Regular season game indicator	Boolean	Flyers, Eagles
ConfQF	Conference quarterfinal indicator	Boolean	Flyers
Start hour	Game start hour	Numeric	All
End hour	Game end hour	Numeric	All
Game period	Pre/during/post game phase	Categorical	All

Lat	Latitude coordinate	Numeric	All
Lng	Longitude coordinate	Numeric	All
Walkoff	Walk-off game indicator	Boolean	Phillies
DefTO	Defensive turnovers	Numeric	Eagles
OffTO	Offensive turnovers	Numeric	Eagles
Division	Division game indicator	Boolean	Eagles
ConfChamp	Conference championship indicator	Boolean	Eagles
SuperBowl	Super Bowl game indicator	Boolean	Eagles

Table 2. Variables used in machine learning models with descriptions and data types.

The four categorical features were converted to binary indicator variables for each category using one-hot encoding. This ensured the dataset was fully numeric and compatible with machine learning algorithms that do not natively support categorical variables.

Each of the four sport datasets was divided into training, validation, and testing subsets to enable a comparison of candidate models using the validation dataset and final model evaluation on the unseen test data. Because the data are aggregated by date, hour, and region, a grouped splitting process was used with day and hour as the grouping variables. This approach ensures that all observations corresponding to the same day and hour remain within a single subset, preventing data leakage across splits. Of the resulting groups, 70% were randomly assigned to the training set, 10% to the validation set, and 20% to the testing set.

The distribution of the target variable, number of police stops, was examined using the training data. Table 3 includes the distribution of hourly police stops in the Eagles' training data. The distribution ranges from 0 to 16 stops per hour, is highly zero-inflated, and is right-skewed.

These observations have important implications for model selection and provide context for interpreting final evaluation metrics.

Number of Stops	Frequency
0	202009
1	9157
2	2748
3	1038
4	469
5	184
6	95
7	88
8 to 16	86

Table 3. Frequency table showing the distribution of hourly police stops per region observed in the Eagles training data.

Model Fitting and Selection

For each sport, candidate models were fit using all combinations across three model classes, two feature sets, and multiple hyperparameter values specific to each model type. The models considered were Random Forest, XGBoost, and LightGBM, chosen for their ability to capture complex, non-linear relationships in large datasets. Random Forest was included due to its robustness to noise and outliers. The gradient boosting algorithms, XGBoost and LightGBM, build sequential trees to iteratively improve model performance, often resulting in strong predictive accuracy. Both models were considered, as LightGBM is computationally efficient for large datasets, while XGBoost utilizes regularization techniques to prevent overfitting.

Exploratory data analysis did not reveal a clear difference in the hourly number of police stops between game days and non-game days (Figures 2, 3, and 4). Therefore, two feature sets were evaluated: one with all available variables and a small set consisting of the non-sport-related features (day of the week, hour, and kilometer-based latitude and longitude coordinates). Hyperparameter tuning was performed by evaluating multiple combinations of parameter values for each model class. In total, 64 candidate models were created for each sport using the training dataset.

Root mean squared error (RMSE) was chosen as the primary evaluation metric for model selection because it assigns greater weight to larger prediction errors compared to the mean absolute error (MAE). This characteristic of RMSE is beneficial given the high proportion of zeros in hourly police stops, as it emphasizes model performance on the rare but important non-zero observations. Lower RMSE values indicate smaller differences between predicted and observed values and therefore a better model.

Each of the 64 candidate models were evaluated using the validation set. To highlight the difference in model performance between feature sets, the RMSE of the top-performing model for each sport and feature set combination is reported in Table 4. All of the top-performing models were LightGBM.

Sport	Feature Set	RMSE
Football (Eagles)	All features	0.404232
	Non-sport features	0.417714
Baseball (Phillies)	All features	0.474663
	Non-sport features	0.486297
Basketball (76ers)	All features	0.456615

	Non-sport features	0.466817
Hockey (Flyers)	All features	0.467042
	Non-sport features	0.480010

Table 4. Model selection results, showing the best RMSE of every sport and feature set combination, evaluated on the validation set.

Across all four sports, models trained using the full feature set had a marginally lower RMSE value than those using only non-sport features, with an average difference of approximately 0.01 stops. Although this difference is small in magnitude, its consistency across sports supports the selection of the full feature set. The chosen final model for each sport was a LightGBM model using all available features and the optimal hyperparameters identified during validation.

Final Model Evaluation

Each final model for the four sports was evaluated using the test set. Additionally, an intercept-only linear model was fit using the training data and evaluated on the test data for each sport. The purpose of these baseline linear models is to serve as a reference point for assessing the performance of the final models.

Both MAE and RMSE were used for final model evaluation. MAE is an easily interpretable measure of average prediction error, while RMSE emphasizes large differences between actual and predicted values. Table 5 shows the MAE and RMSE of the final models and their corresponding baseline models.

	Final Model		Baseline Model	
Sport	MAE	RMSE	MAE	RMSE
Football (Eagles)	0.1046	0.4551	0.1046	0.5328
Baseball (Phillies)	0.1166	0.4692	0.1172	0.5471

Basketball (76ers)	0.1118	0.4572	0.1125	0.5326
Hockey (Flyers)	0.1163	0.4686	0.1178	0.5501

Table 5. Mean absolute error and root mean squared error for the baseline and final model for each sport, evaluated on the testing data.

The MAE values indicate that, on average, both the final model and the baseline models' predictions are off by about 0.11 stops. Given the heavily zero-inflated and right-skewed distribution of the target variable, this similarity suggests that both models are largely influenced by the high frequency of zero-valued observations. The RMSE values for the final models are larger than the corresponding MAE values by approximately 0.35 stops. This discrepancy indicates that the final models make occasional large prediction errors, which are more heavily penalized by RMSE. However, the RMSE values for the final models are consistently lower than those of the baseline models across all sports. This improvement suggests that the final models capture some meaningful variation beyond simply predicting zero hourly stops.

Overall, while the proposed models do not outperform the baseline in terms of average prediction error, they demonstrate improved performance in capturing non-zero observations, as seen by lower RMSE values.

Model Interpretation

Shapley Additive Explanations (SHAP) were used to interpret the final models. For each final model, global feature importance was calculated by computing the mean absolute SHAP value for every feature across a sample of observations from the test data (Molnar n.p.). Across all models, latitude, longitude, and hour of day had the highest importance, with mean absolute SHAP values ranging from 0.05 to 0.08. In contrast, all remaining features had importance values of 0.01 or lower. This pattern indicates that spatial and temporal variables contribute

substantially more to the prediction of hourly police stops than any of the game-related features. These results provide further evidence that game outcome has limited influence on citywide police activity compared to spatial and temporal patterns.

Conclusion

Key Insights

Exploratory analysis revealed no clear differences in the number of police stops between non-game days and single or multiple-game days, nor across the four major Philadelphia sports teams. Similarly, the proportion of stops resulting in arrests or contraband findings was consistent across non-game days and the various game day types. In contrast, clear spatial and temporal trends were observed. Police stop activity increases during the evening, and certain areas of the city exhibit higher stop frequencies than most regions. The temporal pattern aligns with game times between 6 and 8 PM, suggesting a possible relationship. However, this pattern may also reflect broader trends in vehicle, pedestrian, and policing activity. The relatively low stop activity in areas surrounding sports venues highlights a potential data limitation, as stadium-specific security incidents are not included. Overall, these findings suggest that citywide police activity in Philadelphia is driven more by broad spatial and temporal factors than by sporting events.

The modeling results reinforce the conclusions from the exploratory analysis. Machine learning models were developed to predict the frequency of police stops by hour and region, and feature importance techniques were used to identify the main drivers of these predictions. Location and time of day were the most influential variables in making predictions, while all other game-related features were relatively unimportant. These results further support the

conclusion that Philadelphia police activity is influenced primarily by broad spatial and temporal factors rather than game occurrence or outcome.

Impact

While prior literature has analyzed sporting event-related crime across US cities or within stadiums, this study provides a focused view into a city widely known for its intense sports culture. These findings contribute to the broader understanding of how large-scale sporting events influence policing patterns in urban environments. The results suggest that widely held assumptions about Philadelphia's sports culture and its impact on citywide behavior may be overstated. In particular, the analysis indicates that sporting events do not have a measurable effect on police stop frequency or outcomes at the city level. This implies that commonly cited examples of extreme fan behavior in Philadelphia are more likely driven by isolated, highly visible incidents rather than persistent, citywide trends. As a result, this work contributes to ongoing conversations about public safety and the role of policing during major sporting events.

From a data science perspective, this study demonstrates how widely accepted narratives can be evaluated using data visualization and machine learning techniques. The combined approach of exploratory analysis, predictive modeling, and model interpretation provides a framework for assessing whether perceived trends are supported by data. The modeling framework developed in this study could also be used to forecast stop activity, offering a data-driven tool for planning and policy, and reducing reliance on preexisting assumptions. Finally, this study illustrates that null findings, in this case, finding no evidence that sporting events influence policing patterns, can still have a meaningful impact on the field when properly contextualized.

Ethical Considerations & Concerns

Given the sensitive nature of police stop data, this project requires careful attention to ethical implications. Increased police presence during certain days or times may disproportionately affect certain communities or regions, particularly if stop activity is concentrated in specific neighborhoods. While our analysis does not directly measure demographic disparities, the potential for unequal impacts must be acknowledged. Furthermore, predictive modeling in this context raises concerns about reinforcing existing patterns of policing, especially if historical data reflects systemic biases. As we continue this work, it will be important to approach interpretation and application with caution, emphasizing transparency, fairness, and the responsible use of data. Future extensions of this project could incorporate demographic variables to better assess and mitigate potential biases.

We also recognize our results could lead to policy changes that negatively affect the typical game day experience, even for those who are not causing any significant public disruption. For instance, our data exploration results might be used to justify changes in policing strategies around sporting events, even if the observed differences are small or not statistically meaningful. We also hesitate to conclude that an increase in stops or arrests should dictate an increase in police, as an initial increase in police could cause an increase in stops or arrests. Further increasing enforcement could create an overly restrictive atmosphere for fans and residents of Philadelphia and pose a risk of additional unnecessary vehicle and pedestrian stops. Over time, this environment may discourage attendance at sporting events and negatively impact the broader community experience.

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